

The First Fifteen Years of Computing in Aotearoa New Zealand

Brian E. Carpenter, *University of Auckland, New Zealand*

Sathiamoorthy Manoharan, *University of Auckland, New Zealand*

Janet Toland, *Victoria University of Wellington, New Zealand*

Abstract—This article outlines the first fifteen years of modern computing in New Zealand ...

Modern computers arrived in Aotearoa New Zealand in 1960, less than a century after the first discernable information technology [20]. New Zealand was then a relatively isolated and small (2.5 million people) economy, so it is feasible to track the dissemination and socio-economic influence of computing for the following period; this study runs from 1960 to about 1975. Two events in the mid-1970s signaled the end of the economic and social era that began after World War II, and act as the end of our study: Britain joined the European Common Market in 1973, with major economic consequences, and New Zealand switched on a national Police Computer in 1976, with significant social impact.

In 1960, New Zealand had a rather centrally managed economy. Indeed, its currency did not float until as late as 1985. The details are complex [28], but the result was that during the period of our study the Treasury was always concerned about the foreign exchange impact of computer imports, and this necessitated a system of import licensing. This was a constant background for trends in computing, especially since the Treasury initially favored computing service bureaus to ensure maximum usage of what they considered a scarce resource. Of course, many companies much preferred to have their own systems.

Several themes are used to organize this article:

- growth in numbers of computers, vendors and employment
- key areas and types of usage, including service bureaus
- start of computing services and teaching in universities, technical colleges
- professionalization
- local contributions to the technology

- commercial and economic impact
- social impact

GROWTH IN NUMBERS AND EMPLOYMENT

The following table shows the estimated number of electronic digital computers installed in New Zealand over the years in question.

Year	Computers	Source	Notes
1960	2	[19]	
1965	70	[18]	[24] gives 45
1966	81	[18]	
1967	100	[3]	
1968	120	[18]	[12] gives 87
1969	140	[18]	
1971	180	[18]	
1972	200	[12]	
1974	280	[18]	
1976	400	[18]	

A snapshot of installations in Auckland alone lists 37 sites at the end of 1969, covering businesses of many sorts, local government, a university, and even the Naval Research Laboratory [17]. At least seven of the sites appear to be service bureaus. The author noted in passing that there was also a large Government investment in computers in Wellington.

Although there are some discrepancies in the available data, the trend is one of rapid growth, from one computer for every 1.2 million people to one for every 8000. Similarly, the number of programmers in public services rose from a handful (maybe 4) in 1960 to 115 (84 men and 31 women) by 1974 [12]. Overall, 4000 data processing staff were then employed at 220 sites – 39% in Wellington, 31% in Auckland, 9% elsewhere in the North Island, and 21% in the South Island [18]. Pay was attractive: in 1967, an EDP manager's average annual salary was \$4580NZ [3], when the



FIGURE 1. Fletcher Computing Bureau staff, 1966. Courtesy Fletcher Archives, NZ.

national average for company directors and managers was \$4220NZ [5]. Programmers and systems analysts earned anywhere between \$2000 and \$4000.

Even if the public services had 27% of female staff, it is not clear that the same applied elsewhere. For example, a 1966 staff photograph of Fletcher Building's internal computer bureau shows 14 men and only three women (Figure 1). Whenever staff appear in computer room photographs, the women are almost invariably seated at a keyboard and the men are standing in discussion. A 1967 survey [3] shows systems design and programming as 90% male occupations, operations as 50% male, and data preparation as 97% female.

There is little to indicate significant Māori involvement in computing during our period of study. Robyn Kāmira [25] shows that such involvement would only start in the 1980s, when a first handful of Māori information technology specialists graduated.

New Zealand had a long history of preferring to import British products of all kinds, but this did not long survive in the computer market. IBM was of course a major player; not only did they snag the Treasury as their first customer for an IBM 650 in 1960 [19], but they also delivered a heavily discounted IBM 1620 to the University of Canterbury in 1962, the first machine in academia [15]. This was in line with IBM's world-wide practice of educational discounts, a powerful marketing tool that exposed many students to the IBM brand. As time went on, IBM sold numerous machines - more IBM 1620s, at least one IBM 1401, several IBM 1130s, until the game-changing IBM 360 series came along in 1964.

The main British competitor in 1960 was ICT, but a potential customer would certainly have been con-

fused by the fragmented state of the British computer industry at that time [23]. Nevertheless, ICT sold one ICT 1201, at least six ICT 1301s, and numerous machines from the ICT 1900 series. It was not until the various British companies were unified as ICL in 1968 that things settled down, but by then IBM had the lion's share of the New Zealand market and ICL was falling behind. In 1974, the new ICL 2900 series had to compete with the well established and upwards compatible IBM 360 series.

Other companies that were active in the early days were Burroughs (with its E2000, not a general-purpose computer, but specialized for accounting), and NCR (Series 500, also specialized for ledgers).

We have collected available information about individual computers present in New Zealand during the period under study, but at best it is a small sample of the market. At the time of writing, we have identified 81 **[TBD: Check!]** machines (including programmable calculators) with arrival dates up to 1974, but the estimated total in that year was 280. Many of the earlier machines had of course been scrapped by then, so our sample certainly covers less than 25% of the market. The distribution of this sample across manufacturers is shown in the following table. (The total for the British companies that ended up as part of ICL is 29, similar to the total for IBM.)

Manufacturer	Count
Burroughs	12
DEC	9
EELM	3
Elliot	1
HP	1
IBM	28
ICL	8
ICT	17
NCR	2

Throughout the period under study, most computers remained large and comparatively heavy, even as they moved from "first generation" (vacuum tubes) through "second generation" (discrete transistors) and "third generation" (integrated circuits) to the "fourth generation" (very large scale integration). As the technology got smaller, mainframe computers became more powerful rather than more compact; space and air conditioning requirements did not substantially change. For example, the third computer installed in 1969 by Motor Specialties in Auckland, an ICL 1902A, needed to be lifted to the top floor by crane, like its predecessors, an ICT 1201 and an ICT 1301 (Figure 2). Only after 1970 when minicomputers such as the DEC PDP-11 entered the market did this change, so that



FIGURE 2. ICL 1902A Delivery in 1969. Courtesy Fletcher Archives, NZ.



FIGURE 4. Ceramic House, 1969. Courtesy Auckland Libraries Heritage Collections.



FIGURE 3. The TAB Computer Centre, Christchurch, 1970. Courtesy Fletcher Archives, NZ.

some computers could be installed almost as easily as domestic appliances.

Originally, much equipment was rented; in 1967, 70% of “owners” actually rented their equipment, and a further 15% rented part of it [3]. By the late 1960s, the growth in installations led to a minor building boom of dedicated computer centers, which tended to be stark concrete buildings with few windows (Figure 3). A rare exception was Ceramic House in Auckland (Figure 4), specifically designed to accommodate an ICT 1901 delivered in 1969. This building was designated as a heritage site in 2024.

This section has described how computers were imported, housed, installed and staffed during the period of study. We will now consider what they were used for.

HOW COMPUTERS WERE USED

In 1960, the main expected workload was the payrolls for teachers and public servants. The expansion in fields of use came very quickly. By the end of 1966 [10] [TBD: Janet check the Archives citation], Government usage included payroll, statistics, science, NZ Railways, Inland Revenue, Post Office, Social Security, Reserve Bank, Bank of New Zealand, New Zealand Broadcasting Corporation, Apple & Pear Board, and the Dairy Board. In a surviving list of import license applications between August 1965 and May 1966 [10], there are three applications from local government, three from universities, but 30 from business and primary and secondary industries. The New Zealand economy took to computers rather quickly, held back mainly by import license restrictions.

Applications of importance for business users were listed in 1967 as invoicing, payroll, general ledger, stock control, expense analysis and sales statistics [3]. All the same, mathematical, scientific and project planning applications were increasingly used. Users saw the advantages of computerization as speed, accuracy, improved reporting, improved organizational control and staff savings. Quite surprisingly, in 1967 most computers were underutilized, with 250 operating hours per month on average, and single-shift operation still being typical. Only 68% of installations charged internal users for usage: “one of the chief reasons why installations can be uneconomic”.

In 1973, the Department of Statistics provided the following summary [8]:

The demand for computers has come from Government departments, local authorities, universities, primary producer boards, private firms in industries such as printing,

forestry, insurance, oil, food processing, electrical equipment manufacturing, building and construction, clothing, engineering, airways, banking, retailing, motor assembly, paint manufacturing, and stock and station agents.

Because of government's reluctance to issue import licenses, numerous service bureaus arose to allow resource-sharing. By 1967, when the total number of computers was about 100, at least 166 companies were using bureau services operated by computer vendors, and another 78 were using independent bureaus [3]. The following table lists some example bureaus and their approximate dates of foundation, but the list is far from complete.

Year	Name	Location	Notes
1961	Electronic Data Processing Ltd	Auckland	Closed 1963
1963	Electronic Data Systems Ltd	Wellington, later Auckland & Christchurch	50% EELM, 50% NZ Truth (until 1968)
1965	Computer Bureau Ltd	Christchurch, later nationwide	Renamed Datacom
1966	Allied Computer Processors	Dunedin	
1967	Databank Systems Ltd	multiple	Banking consortium
1968	Electronic Data Processing NZ Ltd	Wellington	
1968	NCR Bureau	Wellington	
1968	Burroughs Bureau	Wellington	
1968	ICL Centre	Auckland	
1969	Computer Activities	Auckland	
1969	Computer Systems	Auckland	
1969	Data Enterprises Ltd	Tauranga	Paul and David Walker

The first bureau listed above remains something of an enigma. According to ICT's records in the U.K., it bought an ICT 1301 for "service work"; but we have found no local records to substantiate this, beyond the



FIGURE 5. EDS KDF6, Princes St, Auckland in about 1970. Courtesy Neil Harsant.

formal records of the formation and dissolution of the company. Another phantom bureau hosted the government Education Department's second computer, also an ICT 1301, leased from ICT at the end of 1963 and installed in ICT premises in Wellington. When the Minister of Education pressed a button to inaugurate it, it was said to be "New Zealand's first computer bureau" [2], although all indications are that it was dedicated to the Education Department. It seems that obtaining an import license for a shared bureau computer was significantly easier than for a single-user machine. The other bureaus were real enough, and we now present a few brief case studies.

Electronic Data Systems Ltd (EDS), founded in 1963, was jointly owned by the British company English Electric Leo Marconi (EELM) and the NZ Truth newspaper. It had no connection with the later American company of the same name. EELM was one of the numerous firms that ended up as part of ICL, and presumably wanted to enter the New Zealand market. Apart from its earlier DEUCE and LEO computers, it made a family of KDF machines, including the KDF6, of which at least 12 were sold [23]. The KDF6 sales records are scanty [16], but EDS had such machines in Wellington and Auckland [22]. Indeed we have photographic evidence of the machine in Auckland (Figure 5).

NZ Truth, described by the National Library web site as "one of the country's most colourful, controversial and popular newspapers" ran from 1905 until 2007. It is unclear why it decided to enter the computer bureau business in 1963. In any case, EDS opened a third branch in Christchurch in 1970, and in due course switched to Burroughs and ICL equipment [7]. By 1972, the Christchurch and Wellington offices were linked by

a remote job entry system. EDS remained in business in one form or another until 1994, decades after EELM no longer existed and NZ Truth had sold its stake.

Another success story started in 1965. G. Bernard Battersby (1925-1997) was the Dean of the Faculty of Commerce at the University of Canterbury, and a government advisor. Yet he joined with Paul M. Hargreaves (1938-2014) to found Computer Bureau Ltd (CBL) in Christchurch [14]. Its first computer was an ICL 1902 delivered in 1966; with a staff of seven, CBL was unprofitable until 1969, when a branch was opened in Wellington. In early 1970, CBL had some 50 customers, and the Christchurch computer was working 140 hours per week. Thus an ICL 1902A was delivered to Christchurch, the original ICL 1902 was moved to Wellington, and an additional ICL 1902 was ordered for another new branch in Hamilton. In Auckland, CBL had taken over Fletcher Building's in-house computer bureau, also ICL-based. All training of systems analysts and programmers took place in Christchurch, with total staff expected to reach 70 during 1970 [4]. By 1974, CBL needed to order three ICL 1902T models for Auckland, Wellington and Christchurch, with a nationwide total of six computer centers and 125 staff. Furthermore, they told the newspapers [9]:

[CBL] is unique in that many of its major users are also shareholders, a number of whom previously operated their own inhouse computers. These include Fletcher Holdings, Ltd, Wilson and Horton, Ltd, the New Zealand Co-op Dairy Company, Bing Harris Sargood, Ltd, United Dominion Corporation, Ltd, and in Christchurch, the Canterbury Building Society, the S.I.M.U. Mutual Insurance Association, and the City Council.

Clearly, Computer Bureau Ltd had found a winning formula. In 2025, they still prosper under the name of Datacom, claimed to be the largest technology company in New Zealand, employing at least 6000 people at offices across the Asia-Pacific region.

TBD: short case study of Databank [Janet], ...)

Thus, as computer usage spread throughout business and government, import license pressure encouraged resource sharing and successful bureau operations. This pressure was not to be relieved until major economic reforms in the 1980s, outside our study period.

ACADEMIA AND PROFESSIONALIZATION

Training was an issue from a very early stage. By 1967, it was a serious concern: "It is not sufficient for the

Universities to provide computer facilities and training only at the scientific level. The Technical and Management Institutes cannot be left to their own devices... the secondary schools (and before too long the primary schools) must introduce computer knowledge" [3]. In practice, most training originally came from computer vendors, or people learned on the job. However, by 1970 technical colleges were offering courses in computing and programming, but with only 46 students enrolled nationally [8].

TBD: Start of computing services and teaching in universities, technical colleges (and maybe schools?)

TBD: Start of Computer Science as a discipline

TBD: Origin and growth of NZCS [Janet]

LOCAL CONTRIBUTIONS

TBD: Local contributions to the technology [Mano]

In 1966, Bruce Moon (then at the University of Canterbury) published the first local programming textbook [26].

In 1968, Progeni, claimed to be the first software house in New Zealand, was founded by Perce Harpham (1932-2xxx) [21].

[Brian's anecdote 1:] In 1974, with increasing demand for interactive computing, Massey University decided that Burroughs block-mode terminals were too expensive. Instead, the Computer Unit bought a DEC PDP 11/10 to act as a terminal concentrator for inexpensive DEC VT05 terminals. Lacking software, the only answer was DIY. Brian Carpenter converted CERN's compiler for the PL-11 programming language to run on the Burroughs mainframe, and Ted Drawneek wrote the necessary software.

[Brian's anecdote 2:] In 1975, teams at Massey University and Victoria University of Wellington launched a project called Kiwinet, whose initial goal was resource sharing between the B6700s at the two universities. Despite establishing connectivity, there was no software ecosystem and the project was mainly a learning experience, as discussed in Chapter 2 of [27]. Internet connectivity only came to New Zealand in 1989.

ECONOMIC IMPACT

In 1960, the New Zealand economy was very tightly coupled to the UK, despite being literally a world away. The New Zealand currency was held at par with the British pound, and the UK accounted for 53% of exports and 44% of imports. Moreover, 90% of the total value of exports was derived from agricultural produce [1]. Nevertheless, diversification had started, with meat exports going to 60 different countries. Thus, by 1975 exports to the UK were dramatically down to 22% of the total and imports to 19% [13].

Britain had first applied unsuccessfully for admission to the European Economic Community (EEC) in 1963, and this was correctly interpreted in New Zealand as a serious warning that access to the UK market would sooner or later become difficult. There was a successful effort at diversification of both exports and imports. As noted above, IBM and other American computer companies took advantage of this opportunity. Although New Zealand's access to UK markets was eased by a so-called Luxembourg agreement in 1971, when Britain eventually joined the EEC in 1973, New Zealand had already become a completely independent trading nation, living on its wits, as it remains today.

Thus the growth of computing in New Zealand took place against a background of major economic upheaval....

SOCIAL IMPACT

Perhaps it is no surprise that lawyers were among the first to consider the social impact of computers. A first legal analysis was published in 1971 by Francis M. Auburn [17], in which consideration of privacy was the major theme. The author observed that "there is a fear of computers and their operators among the general public" and quoted a rather naive counter-argument that "invasions of privacy are the result of misuses of the data retrieved and not a function of the storage medium" (i.e., blame the users, not the technology). With hundreds of thousands of banking transactions per day already taking place, and with a national police computer and computerized hospital records in the offing, the risks of accidental or intentional privacy breaches already seemed very real, more than 50 years ago. The legal situation was murky: "If there is a general right of privacy we are still no nearer to a definition of the circumstances in which it is applicable to computers." All sorts of data raised thorny privacy issues once computerised, such as medical records, census returns, and credit history. The issue of a unique national identity number to tie all records together was also of concern, with inevitable references to George Orwell's novel *1984*. Auburn commented that "The most rigorous presently known technological safeguards would be quite insufficient to prevent abuse." Auburn's conclusion was that a "comprehensive review of the present and future impact of electronic data processing on society" was needed, to be followed by legislation.

Auburn was not alone in raising these concerns. For example, also in 1971, A. L. C. Humphreys of ICL, the Christchurch *Press*, and the New Zealand Computer Society were all concerned about privacy.



FIGURE 6. Wanganui Computer Centre, 1975. Courtesy Fletcher Archives, NZ.

Certain lawyers even drafted a proposed law [6] and a Bill was introduced to Parliament in 1975 [11] but never made it into law. New Zealand's first Privacy Act was not passed until 1993, prior to which common law applied.

Despite this legal vacuum, plans for a national police database were made from 1972. The system, officially the National Law Enforcement Data Base, but often called the Wanganui Computer after the town now known as Whanganui, was operational by 1976 (Figure 6). This led some years later to protest actions. However, it seems that during our study period, despite the concerns of some lawyers, there was little public perception of computers as a societal threat.

Apart from privacy issues, it is hard to separate the impact of computers from other factors that changed society in the 1960s and early 1970s, as the first generation born after World War II grew to maturity. Beardon [18] discusses the computerization of "women's work" and its possible impact on unemployment. Certainly, the many skilled women who operated ledger machines and comptometers found themselves out of a job as computers took over. The replacement jobs, if any, would have been less skilled and lower paid data entry jobs. Beardon noted that the fraction of the "unemployed who are female" increased from 10.9% in 1961 to 18.0% in 1971 and 34.9% in 1975, but is impossible to know how much of this change was caused by computers. The loss of typing and secretarial jobs to computers came much later, as did the availability of home computers. Throughout our period of study, computers were mainly hidden from public view except for people directly involved, and had relatively little social impact compared to other modern technology.

such as television. Apart from the job market, where computers had a direct effect on some working lives, they were not even a talking point.

CONCLUSION

TBD

ACKNOWLEDGMENTS

TBD

BIBLIOGRAPHY

- [1] *The New Zealand Official Year-Book, 1962*. Statistics New Zealand, 1962. [Online]. Available: https://www3.stats.govt.nz/new_zealand_official_yearbooks/1962/
- [2] "Computer will process School Cert. Results," *National Education - The Journal of the New Zealand Educational Institute*, vol. 45, no. 494, p. 506, December 1963.
- [3] *New Zealand Computer Processing - Second annual survey of New Zealand Electronic Data Processing*. W. D. Scott & Company, 1967.
- [4] "New Computer Is Worth \$500,000," *The Press (Christchurch)*, p. 12, January 16 1970.
- [5] *The New Zealand Official Year-Book, 1970*. Statistics New Zealand, 1970. [Online]. Available: https://www3.stats.govt.nz/new_zealand_official_yearbooks/1970/
- [6] "Bill prepared to protect privacy," *The Press (Christchurch)*, p. 23, June 19 1971.
- [7] "Computer Link for Christchurch," *The Press (Christchurch)*, p. 12, March 8 1972.
- [8] *The New Zealand Official Year-Book, 1973*. Statistics New Zealand, 1973. [Online]. Available: https://www3.stats.govt.nz/new_zealand_official_yearbooks/1973/
- [9] "Heavy investment by Computer Bureau," *The Press (Christchurch)*, p. 13, August 24 1974.
- [10] "Machines – other departments – Computers – electronic & customs import licences, NZ Archives Reference No AALR W31158 873 Box 4," 1974.
- [11] "Privacy bill heard," *The Press (Christchurch)*, p. 2, July 26 1975.
- [12] *The New Zealand Official Year-Book, 1975*. Statistics New Zealand, 1975. [Online]. Available: https://www3.stats.govt.nz/new_zealand_official_yearbooks/1975/
- [13] *The New Zealand Official Year-Book, 1977*. Statistics New Zealand, 1977. [Online]. Available: https://www3.stats.govt.nz/new_zealand_official_yearbooks/1977/
- [14] "A few of the first," in *Looking Back to Tomorrow*, W. R. Williams, Ed. The New Zealand Computer Society, 1985. [Online]. Available: <https://history.itp.nz/part-3/a-few-of-the-first.html>
- [15] "History of the Computer Services Centre," University of Canterbury, Tech. Rep., 2015.
- [16] "List of probable deliveries of English Electric KDN2, KDF6 and KDF7 computers," 2022. [Online]. Available: <https://www.ourcomputerheritage.org/Maincomp/Eel/ccs-n2x1.pdf>
- [17] F. M. Auburn, "Legal Problems of Storage and Processing of Electronic Data," in *New Zealand Legal Research Foundation Seminar Papers 6: Computers and the Law*. New Zealand Legal Information Institute, 1971, ch. 5, pp. 108–121. [Online]. Available: <https://www.nzlii.org/nz/journals/NZLRFSP/1971/6.html>
- [18] C. Beardon, *Computer Culture - The Information Revolution in New Zealand*. Reed Methuen, 1985.
- [19] B. E. Carpenter, "The First Computer in New Zealand," *IEEE Annals of the History of Computing*, vol. 42, no. 2, pp. 33–41, 2020.
- [20] B. E. Carpenter and S. Manoharan, "Information Technology Pioneers of Aotearoa New Zealand," *IEEE Annals of the History of Computing*, vol. 47, no. 2, pp. 44–58, 2025.
- [21] P. Harpham, "Progeni, 1968–89: Success and failure," in *Return to Tomorrow: 50 Years of Computing in New Zealand*, J. Toland, Ed. The New Zealand Computer Society, 2010. [Online]. Available: <https://history.itp.nz/part-2/harpham.html>
- [22] N. Harsant, Private communication, 2025.
- [23] J. Hendry, *Innovating for Failure: Government Policy and the Early British Computer Industry*. MIT Press, 1989.
- [24] Hone Heke (pseudonym), "Computing in New Zealand," *Datamation*, vol. 11, no. 3, pp. 41–42, March 1965.
- [25] R. Kāmira, "Te Hikoi Roa: A journey of Māori towards information technology mastery," in *From Yesterday to Tomorrow - 60 years of tech in New Zealand*, J. Toland, Ed. IT Professionals New Zealand, 2021, ch. 13. [Online]. Available: <https://history.itp.nz/part-1/kamira.html>
- [26] B. A. M. Moon, *Computer Programming for Science and Engineering*. Butterworths, 1966.
- [27] K. Newman, *Connecting the Clouds - the Internet in New Zealand*. Activity Press, 2008. [Online]. Available: <https://www.nethistory.co.nz/>
- [28] Sullivan, Richard, "Exchange rate fluctuations: How has the regime mattered?" *Reserve Bank of New Zealand: Bulletin*, vol. 76, no. 2, pp. 26–34, 2013.

Brian E. Carpenter is an Honorary Professor in the School of Computer Science, University of Auckland, New Zealand. He is interested in Internet protocol design and in computing history. He received a Ph.D. from the University of Manchester, U.K. and is a past Chair of the Internet Engineering Task Force. Contact: brian@cs.auckland.ac.nz

Sathiamoorthy Manoharan is a Senior Lecturer in the School of Computer Science, University of Auckland, New Zealand. He is a senior member of IEEE. He is interested in computer systems, particularly with respect to performance and security. Contact: mano.manoharan@auckland.ac.nz.

Janet Toland is currently an Associate Professor in the School of Information Management, Te Herenga Waka, Victoria University of Wellington, Wellington, New Zealand. Her research work is in the history of information systems. She has a particular interest in social and ethical issues. She is currently the Historian for the Association of Information Systems. Contact her at Janet.Toland@vuw.ac.nz.